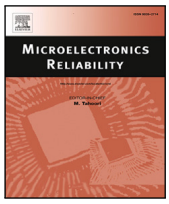
Microelectronics Reliability 182 (2026) 116158



Contents lists available at ScienceDirect

Microelectronics Reliability

journal homepage: www.elsevier.com/locate/microrel



Research paper



Random-vibration fatigue life of plated through-hole components on circulator pump PCBs: Experiments, model fidelity and spectral damage methods

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ARTICLE INFO

Keywords:
Random vibration
Printed circuit board
Plated-through-hole components
Finite element modelling
Fatigue life prediction

ABSTRACT

This study combines durability testing with predictions of its results obtained using a controlled hierarchy of finite-element (FE) models for heavy plated through-hole (PTH) components mounted on a circulator-pump Printed Circuit Board (PCB). Two operation-critical heavy PTH components on a circulator-pump PCB were subjected to dedicated broadband random-vibration durability experiments to establish benchmark fatigue lives, which then served as the reference for evaluating frequency-domain fatigue-life predictions. Predictions from simplified component idealisations with nominal damping did not match the measured lives, motivating FE lead refinement and updated damping. Mode-specific damping ratios were obtained from displacement measurements acquired by Digital Image Correlation (DIC). Fatigue life was estimated from lead-bend hotspot equivalent von Mises stress power spectral densities (PSDs) using Dirlik's spectral method and Steinberg's three-band method and compared with measured time-to-failure. Using locally refined lead geometry together with component's mode-specific damping values substantially improved agreement in the high-excitation cases: Dirlik-based predictions matched the experimental mean lives within approximately a factor of two, whereas Steinberg's method remained strongly nonconservative. The results show that accurate random-vibration life prediction for heavy PTH components requires targeted local lead fidelity and experimentally identified damping, and that spectral fatigue methods provide more reliable estimates than empirical three-band rules for lead-bending driven failures.

1. Introduction

Broadband random excitation commonly encountered in machinery-driven electronics causes high-cycle fatigue in PCB assemblies through repeated bending and local deformation at component terminations, and in circulator pumps this vibration (from pressure pulsations, rotating elements, and mounting transmission) is well represented as a stationary acceleration power spectral density (PSD) [1]. Random-vibration fatigue experiments reported in the electronics literature focus predominantly on surface-mount technology (SMT) interconnects [2–6]. Failure is commonly defined and monitored through *in-situ* electrical continuity measurements using daisy-chain circuits, often aligned with IPC-9701A-style event or data-logger criteria based on resistance excursions and sustained increases [7].

Through-hole technology (THT) components, including plated through-hole (PTH) parts used for power conditioning and surge protection, shift the dominant vibration-fatigue mechanism away from an SMT solder joint and toward bending of the lead wires and stress concentrations near the lead knee and the lead–solder transition. In this class of components, the comparatively stiff body acts as an inertial mass, and cyclic relative motion between the body and the PCB is transferred into alternating lead bending, making the lead and its immediate attachment region natural crack-initiation sites. Experimental work on leaded components under random vibration has confirmed that failures often localise in the lead/solder region rather than in the component body, and has used test data to support physics-based interpretation of the failure process [8]. At the same time, a substantial fraction of the random-vibration fatigue literature in electronics continues to treat solder joints as the primary failure location and develops fatigue-damage estimation workflows around solder-joint stress PSDs, including probabilistic extensions [9]. This reveals a clear gap: compared with the SMT daisy-chain evidence base, relatively few broadband random-vibration benchmark experimental data (i) isolate lead-dominated PTH failures,

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https://doi.org/10.1016/j.microrel.2026.116158
Received 10 February 2026; Received in revised form 25 March 2026; Accepted 27 April 2026
Available online 8 May 2026
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Nomenclature

$S_{eq}(f)$	Equivalent von Mises stress PSD [MPa^2/Hz]
S_{vm}	RMS equivalent von Mises stress amplitude at hotspot (1σ) [MPa]
m_n	nth spectral moment of $S_{eq}(f)$
ν_p	Peak rate of the equivalent stress process
$\dot{D}$	Fatigue damage rate (damage per unit time)
D	Accumulated fatigue damage
T	Exposure time
A, b	Basquin parameters in the stress–life relation $N = A\,S^{-b}$
N_0^+	Zero-crossing rate [Hz]
$P_{a,i}$	Input acceleration PSD level at mode i [g^2/Hz]
P_i	Probability weight of the ith Gaussian stress band in Steinberg's method
Ω_i	circular frequency [rad/s] of mode i
ξ_i	Modal damping ratio of mode i
Q_i	Transmissibility factor, $Q_i = 1/(2\xi_i)$
Δf_i	Modal bandwidth, $\Delta f_i = f_i/Q_i$
N_2, S_a	Reference cycles-to-failure and reference stress amplitude
G	Three-band stress scale factor
$\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$	Normal stress components
$\tau_{xy}, \tau_{yz}, \tau_{xz}$	Shear stress components
E	Error factor, $E = T_{\text{model}}/T_{\text{exp}}$
S_1	First-order Sobol sensitivity index
S_T	Total-effect Sobol sensitivity index

(ii) report time-to-failure using clearly defined failure criteria, and (iii) provide the stress/response measurements and failure localisation needed to support or validate lead-stress-based fatigue predictions.

To predict fatigue life of PTH leads, the FE model must reproduce the PCB boundary conditions and the local compliance at the component attachment so that the simulated stress response can be compared to the experiment. For the PCB, a dominant simplification in vibration analysis remains the smeared property approach, where the multilayer stack-up is homogenised into an equivalent plate and represented with effective, often isotropic, properties calibrated from simple bend or modal tests [10,11]. Recent work has shown that multilayer PCBs can also be represented using more advanced multiscale and partitioned homogenisation approaches that retain directional stiffness information more accurately for heterogeneous conductive-layer layouts [12,13]. The present study is concerned with the effect of lead-model fidelity on the predicted stress and fatigue life. This can be examined using a globally isotropic-equivalent PCB model, without the additional complexity of a fully anisotropic board model. For components, practical assembly models often rely on lumped masses for bodies, beam-like lead representations, and idealised solder geometries, sometimes supplemented by local submodels where detailed stresses are required [14–16]. Commercial workflows such as those implemented in Ansys Sherlock [17] typically rely on simplified component representations. These assumptions are efficient, but the magnitude of error introduced by common PCB/lead simplifications at the local level—where lead stress is evaluated for fatigue—remains insufficiently quantified. Further, this uncertainty is compounded by damping: standard simulations often apply a constant damping ratio (e.g., 1%–2%) across the frequency range, despite the frequency-dependent dissipation behaviour of populated boards, leaving limited clarity on how generic damping assumptions bias predicted lead stresses [18].

In engineering practice for electronics, Steinberg's displacement-based guideline and the related three-band damage method remain widely used because they provide a simple route from a prescribed random-vibration environment to a durability assessment [19–21]. For vibration-driven failures of metallic leads and interconnects, the damage mechanism is typically high-cycle fatigue (HCF), so life is commonly predicted using a stress–life (S–N/Basquin-type) relationship together with Miner's rule [22]. In the three-band formulation, the random response is treated as Gaussian and discretised into three amplitude levels (1σ, 2σ, and 3σ), which are combined with an estimated cycle rate to obtain accumulated damage with minimal modelling effort [19,20]. That convenience comes with limitations due to empirical calibration, and test-informed studies have reported systematic deviations and motivated local strain-based formulations [23,24].

Frequency-domain spectral fatigue methods provide an alternative, estimating damage from the stress/strain PSD via analytical approximations of rainflow statistics; a widely used choice is Dirlik's method [9,25]. Reported comparisons indicate that predicted damage can shift with the damage-model choice, particularly for broadband or multi-peaked responses [26]. For heavy PTH components, it therefore remains unclear which approach yields consistent predictions, given that assumptions underlying Steinberg's method may be violated by local component rocking and that spectral methods depend strongly on the accuracy of the local stress input.

This work develops a traceable workflow for random-vibration fatigue of heavy PTH components in a machinery environment by combining durability testing, a hierarchy of FE models with controlled simplifications, and test-calibrated damping identification. In the present workflow, a uniform damping assumption is used first as a practical baseline to obtain a finite linear response, after which the predictive calculations are carried out using mode-specific damping values identified for the PCB and the two critical components.

The paper is organised as follows. Section 2 presents the random-vibration durability experiments, including the failure definition and monitoring approach used to obtain time-to-failure data. Section 3 introduces the two fatigue-life estimation methods used in this work and clarifies the stress inputs they require, which are obtained from finite element analysis. Section 4 describes the FE modelling strategy and model hierarchy and reports the extracted lead stress responses. Section 5 compares the resulting fatigue-life predictions with the experimental observations and discusses the influence of model fidelity, damping, and fatigue parameters. Section 6 concludes the paper.

2. Experimental methods

2.1. Durability test setup

The PCB assemblies were mounted on an electrodynamic shaker and subjected to broadband random base excitation in the frequency range 100–1700 Hz. A total of ten nominally identical PCBs were tested under two excitation cases, with five repetitions per case to quantify repeatability while comparing two orthogonal base-excitation directions representative of the in-situ mounting orientation:

- Case X–0.20: random excitation in the X direction with a constant input PSD of 0.20 g^2/Hz, corresponding to an overall root-mean-square (RMS) acceleration level of 17.9 G_{RMS}.
- Case Y–0.05: random excitation in the Y direction with a constant PSD of 0.05 g^2/Hz, corresponding to 8.95 G_{RMS}.

These excitation levels were defined as an accelerated random-vibration profile comparable in severity to MIL-STD-202G, Method 214 A (X–0.20 ≈ Condition E and Y–0.05 between Conditions B–C), and Steinberg's equivalence guideline were estimated to represent ~15 years of field exposure at 1.5 G_{RMS} [19,27].

The PCB assemblies were mounted on a portable electrodynamic shaker, as illustrated in Fig. 1. A random excitation signal generated

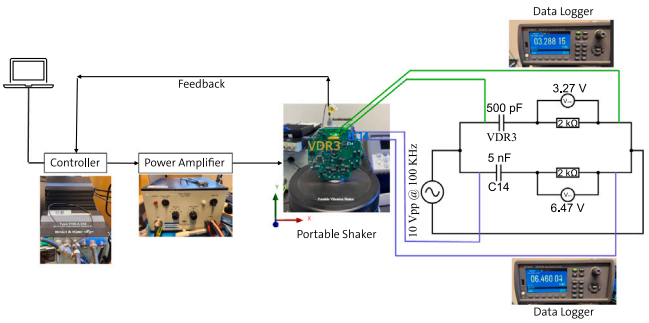


Fig. 1. Durability and electrical monitoring setup for the random-vibration fatigue tests.

by a vibration controller was fed to a power amplifier, which supplied the current required to drive the shaker. A control accelerometer attached to the shaker table provided feedback to the controller to maintain the specified PSD level and compensate for any drift in the excitation during the test. This closed-loop acceleration control approach is standard practice in laboratory vibration testing and is consistent with established environmental vibration test procedures (e.g., control-accelerometer instrumentation and control schemes in MIL-STD-810H) [28].

2.2. Failure monitoring and life definition

The time to failure of the through-hole components was recorded using an electrical monitoring circuit, schematically indicated in Fig. 1. A sinusoidal monitoring signal of 10 V_{pp} at 100 kHz was applied simultaneously to components VDR3 and C14. VDR3 is a metal-oxide varistor used for surge protection; under the low-amplitude monitoring excitation (well below its clamping voltage) it remains in its high-impedance state, and the measured terminal response is dominated by its small parasitic capacitance. This provides a continuously measurable electrical quantity that is lost when a lead fractures, enabling continuity-based life recording for VDR3 in the same manner as for the capacitor C14. To enable continuous monitoring, each component (VDR3 and C14) was connected in parallel with a 2 kΩ resistor, and the RMS voltage across the component was recorded. The two voltages were acquired on separate channels of the data logger throughout the random-vibration test. While in-situ electrical monitoring is common in electronics reliability testing, published random-vibration studies on lead-dominated PTH failures rarely report the specific monitoring circuit; the present description is therefore included for reproducibility. Monitored voltage histories for individual tests are shown in Figs. 2 and 3. As long as both leads remain intact, the RMS voltage across the component stays approximately constant. The first drop of the monitored voltage to (or very close to) zero typically occurs when one of the leads fractures through the thickness and a local separation forms. After this instant, intermittent mechanical contact between the fractured surfaces can temporarily re-close the circuit, causing the voltage to return briefly to its nominal level. In this study, the fatigue life of each component is defined as the elapsed test time at which the first zero-voltage event is observed in the monitored signal. Fig. 4 shows representative before- and after-test images; for both components C14 and VDR3, fracture of one of the leads is clearly visible in the post-test condition.

2.3. Durability test matrix

The random-vibration durability programme and measured times to failure are summarised in Table 2.1. Five nominally identical PCB assemblies were tested for each excitation case (A1–A5 for X–0.20 and A6–A10 for Y–0.05).

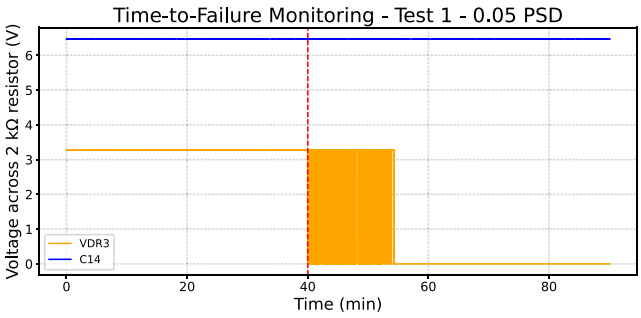


Fig. 2. Voltage history for a PCB durability test Y-0.05.

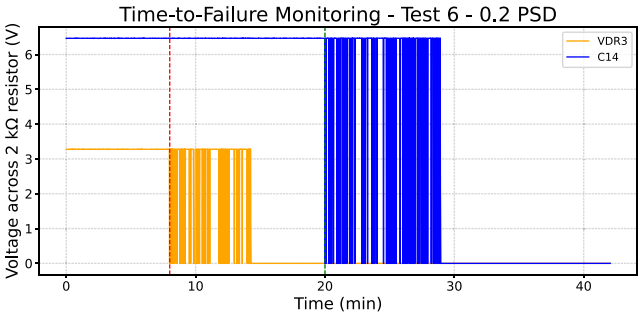


Fig. 3. Voltage history for a PCB durability test X-0.2.

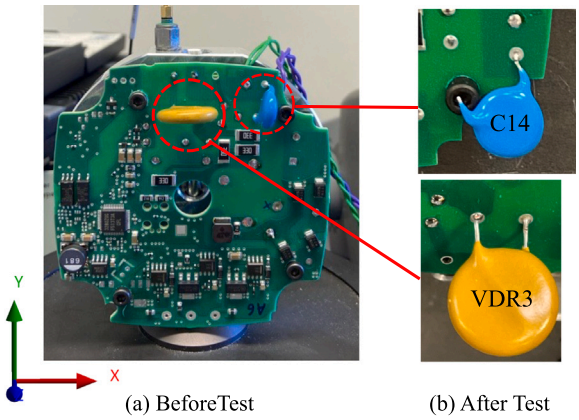


Fig. 4. C14 and VDR3 before and after random-vibration testing.

For each PCB, the time to failure was recorded for VDR3 and C14 using the criterion defined in Section 2.2. Under the higher X–0.20 loading, both VDR3 and C14 failed in all five tests, with mean times to failure of 9.0 min and 19.6 min and sample standard deviations of approximately 2.2 min and 2.7 min, respectively. This level of scatter indicates reasonably repeatable results under nominally identical conditions. Under the milder Y–0.05 loading, only VDR3 failed within the test duration; its mean time to failure was 34.6 min with a sample standard deviation of about 4.3 min. No failures of C14 were observed for A6–A10. Post-test inspections across all failed specimens and both loading cases showed that cracking consistently localised in the lead close to the solder-joint side, identifying this region as the critical location for lead-dominated failures in the present assemblies. Each durability test was terminated after a maximum duration of 1.5 h. This upper limit was chosen to keep the overall test campaign within practical time and equipment constraints; components that had not failed by this time were treated as run-outs. The chosen repetition count (five specimens per loading case) was confirmed to be sufficient

Table 2.1
Durability test results: individual times to failure and mean ± sample standard deviation.

PCB ID	PSD (g^2/Hz)	Time (min.)		Avg. ± SD (min.)	
		VDR3	C14	VDR3	C14
A1		8	20		
A2		10	18		
A3	X–0.20	9	23	9.0 ± 2.2	19.6 ± 2.7
A4		6	16		
A5		12	21		
A6		40	–		
A7		35	–		
A8	Y–0.05	29	–	34.6 ± 4.3	–
A9		32	–		
A10		37	–		

for characterising the mean durability response, as indicated by the observed scatter in Table 2.1. The average experimental times to failure reported in Table 2.1 are used in the following sections as the reference for validating the numerical fatigue-life predictions.

3. Fatigue damage methods

In this work, fatigue life under broadband random excitation is evaluated using two frequency-domain damage models: Dirlik's spectral method (Section 3.1) and Steinberg's three-band method (Section 3.2). For fatigue assessment, the stochastic stress response is described in the frequency domain by PSDs, and the scalar stress process used for the damage calculations is the equivalent von Mises stress process with PSD $S_{eq}(f)$ defined following Preumont et al. [29]. The stress response required to construct $S_{eq}(f)$ is obtained from the random-vibration analysis in ANSYS [30], whereas the fatigue parameters are selected within the statistical bounds for annealed copper wire reported by Harlow [31].

3.1. Dirlik spectral method

The Dirlik spectral method [32] is used to estimate fatigue damage directly from the equivalent von Mises stress PSD $S_{eq}(f)$. For a one-sided stress PSD $S_{eq}(f)$, the nth spectral moment is

$$m_n = \int_0^\infty (2\pi f)^n \, S_{eq}(f)\, \mathrm{d}f, \tag{1}$$

and the associated peak rate of the equivalent stress process is

$$\nu_p = \sqrt{\frac{m_4}{m_2}}. \tag{2}$$

Dirlik's formulation yields the damage rate (damage per unit time) in closed form as

$$\dot{D}_{\mathrm{Dirlik}} = \frac{\nu_p}{A} (2\sqrt{m_0})^b \left[D_1 \frac{\Gamma(1+b)}{Q^b} + (\sqrt{2})^b \, \Gamma\left(1+\tfrac{b}{2}\right) \times \left(D_2 |R|^b + D_3\right) \right], \tag{3}$$

where $\Gamma(\cdot)$ is the Gamma function, and D_1, D_2, D_3, Q, and R are dimensionless coefficients expressed in terms of the spectral moments m_0, m_1, m_2, and m_4 according to the standard Dirlik relations [32]. The accumulated damage over an exposure time T is then

$$D_{\mathrm{Dirlik}} = \dot{D}_{\mathrm{Dirlik}} \, T. \tag{4}$$

Inputs from FE (ANSYS): stress-component PSDs at the critical lead location are exported from ANSYS and used to form $S_{eq}(f)$; the spectral moments m_n (Eq. (1)) and the damage rate $\dot{D}_{\mathrm{Dirlik}}$ (Eq. (3)) are evaluated in Python from these exported PSDs. **Inputs from the S–N curve:** the fatigue parameters (A, b) in Eq. (3) are taken from the copper-lead stress–life relation adopted in this work.

3.2. Steinberg's three-band method

The Steinberg fatigue model [19] is an empirical approach widely used to estimate the fatigue life of electronic assemblies subjected to random vibration. The average number of stress cycles per second is characterised by the zero-crossing rate N_0^+, which is evaluated using Rice's formulation [33] as

$$N_0^+ = \frac{1}{2\pi} \left(\frac{\sum_i \dfrac{P_{a,i}\, \Delta f_i \, Q_i^2}{\Omega_i^2}}{\sum_i \dfrac{P_{a,i}\, \Delta f_i \, Q_i^2}{\Omega_i^4}} \right)^{1/2} \quad [\mathrm{Hz}], \tag{5}$$

where $P_{a,i}$ is the input acceleration PSD level (g^2/Hz) at the natural frequency f_i, $\Omega_i = 2\pi f_i$ is the circular frequency, $Q_i = 1/(2\xi_i)$ is the transmissibility factor at the ith resonant frequency, ξ_i is the modal damping ratio, and $\Delta f_i = f_i/Q_i$ is the bandwidth associated with each resonance. In the present workflow, f_i are taken from the FE modal, while $P_{a,i}$ and ξ_i follow directly from the prescribed random-vibration environment and the damping description applied in the FE analysis; together these FE-defined quantities provide the inputs required to evaluate N_0^+.

The damage rate is written in the compact form

$$\dot{D}_{\mathrm{Steinberg}} = \frac{N_0^+}{N_2 S_a^b} \, (G\, S_{vm})^b, \tag{6}$$

where

$$G = \left(\sum_{i=1}^{3} P_i \, i^b \right)^{\frac{1}{b}}. \tag{7}$$

Inputs from FE (ANSYS): ANSYS is used to compute the element-level equivalent von Mises stress response under random-vibration excitation, and the RMS equivalent von Mises stress amplitude, S_{vm}, is extracted at the critical lead location from the ANSYS random-vibration solution. **Inputs from the S–N curve:** the parameters (N_2, S_a, b) are taken from the copper-lead stress–life relation adopted in this work.

4. FE modelling strategy

This section describes the finite element (FE) modelling workflow used to predict the random-vibration stress response and fatigue life of plated through-hole (PTH) component leads on the circulator pump PCB. All models are implemented in *ANSYS Mechanical* and analysed with identical fixture constraints and base-excitation definitions so that observed differences in stresses and predicted life can be attributed to modelling assumptions (PCB representation and lead-detail level), rather than boundary conditions.

4.1. PCB model hierarchy

A hierarchy of PCB-level FE models was constructed to isolate the sensitivity of fatigue predictions to model fidelity, specifically comparing homogenised representations against explicit geometric details (Fig. 5-I and III). The high-fidelity reference, **M0**, represents the PCB as a seven-layer solid stack with four copper-bearing signal layers and three FR-4 dielectric laminates arranged according to the actual stack-up. The copper-bearing layers are modelled as reinforcement traces within the layered section, while the fixture and component population follow the assembled hardware. This model provides the most direct representation of the physical stack-up, but its large number of degrees of freedom makes it impractical for iterative design studies and repeated fatigue-life evaluations under multiple load cases.

To enable efficient random-vibration stress and fatigue computations, two simplified global PCB representations were used:

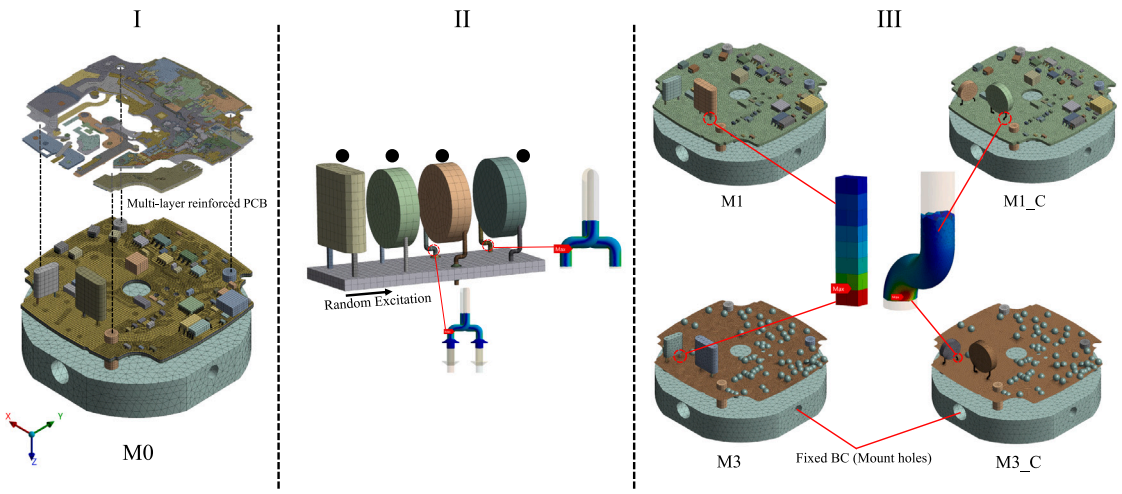


Fig. 5. Overview of the modelling workflow. **I:** M0 layered reference PCB. **II:** Lead Configurations 1–4. **III:** PCB-level models M1 and M3, with refined M1_C/M3_C where VDR3 or C14 uses the selected detailed lead model.

- **M1 (homogenised solid PCB):** the multilayer stack of M0 is replaced by a single solid body with equivalent isotropic properties (Section 4.3), while the fixture and component placement remain unchanged.
- **M3 (shell PCB with lumped components):** the PCB is represented by shell elements with the same homogenised properties as M1, and the non-critical components are represented as lumped point masses located at their centroids and connected to the PCB via rigid multi-point constraint (MPC) equations.

Finally, for both global PCB representations, refined variants were introduced: M1_C and M3_C replace the original representation of the critical PTH component (VDR3 or C14) with a detailed lead geometry selected from the dedicated single-component lead study (Section 4.2). This isolates the impact of lead-detail level on hotspot stresses and fatigue-life predictions while maintaining the same global PCB description.

4.2. Lead modelling and hotspot selection

Sherlock-based workflows typically employ simplified lead representations that are computationally efficient but may not resolve the local lead-bend geometry where vibration-induced fatigue damage initiates. To address this limitation, a set of dedicated single-component FE models was developed (Fig. 5-II). Four lead configurations (1–4) were considered, with increasing geometric detail in the lead-bend and solder regions. In all cases, the distance between the component disc centroid and the PCB surface, the lead-bend angle, and the free lead length were taken from measurements on the physical components to obtain representative lead geometries. Hotspot selection was guided by the experimental failure observations. Post-test inspections consistently showed crack localisation in the lead on the solder-joint side, close to the lead–solder transition (Section 2.3), indicating that the fatigue-critical region is the lead bend immediately above the PCB and adjacent to the solder fillet.

The subsequent PCB-level fatigue analysis focuses on Configurations 3 and 4, because these explicitly resolve the lead bend above the PCB, the location consistently observed to govern the maximum equivalent stress under random excitation. Configuration 3 includes the solder fillet together with the lead, whereas Configuration 4 omits an explicit solder geometry and connects the lead directly to the PCB surface. Under identical excitation, both configurations predict the same critical hotspot at the lead bend, and the peak equivalent stress differs by only ~1.5%, indicating that explicit solder modelling has negligible influence on the governing lead-bend stress. Therefore Configuration 4 was adopted for the PCB-level refined models M1_C and M3_C (Fig. 5-III). In the fatigue calculations, stresses are evaluated at the lead-bend hotspot rather than at the lead–PCB connection, thereby avoiding interpretation issues associated with junction singularities.

4.3. Material properties

The materials are modelled as linear elastic. The reference PCB model M0 uses the layer-wise seven-layer stack-up listed in Table 4.1. In the simplified models M1, M1_C, M3, and M3_C, the PCB is replaced by an equivalent isotropic representation so that the global board response can be retained in a computationally efficient form for the controlled model-fidelity comparison. The same table also lists the copper-lead properties used for VDR3 and C14. In M0, the conductive signal layers are represented using effective properties based on their copper coverage, whereas the dielectric layers use the Generic FR-4 laminate properties. In M1 and M3, the multilayer stack-up is reduced to an equivalent isotropic PCB using a smeared-property approach. The effective Young's modulus E_{eq} and density ρ_{eq} are computed as volume-weighted averages of the constituent layers,

$$E_{\mathrm{eq}} = \sum_{i=1}^{n} V_i E_i, \qquad \rho_{\mathrm{eq}} = \sum_{i=1}^{n} V_i \rho_i, \tag{8}$$

where V_i, E_i, and ρ_i are the volume fraction, Young's modulus, and density of layer i, respectively. The resulting equivalent PCB properties are reported in Table 4.1. This isotropic reduction is used only for the simplified global PCB models and does not reproduce the full directional behaviour of the multilayer board. For this representation, $\nu_{\mathrm{eq}} = 0.14$ is adopted in models M1 and M3.

4.4. Meshing, boundary conditions, and baseline damping assumption

Model M1 is discretised with 3D solid elements (`SOLID186`) for the PCB, whereas model M3 employs shell elements (`SHELL181`) for the PCB with non-critical components represented as lumped point masses connected via rigid MPC equations. The refined variants (M1_C and M3_C) replace the critical component (VDR3 or C14) with the selected detailed lead model (Configuration 4) and include local mesh refinement in the lead-bend region. Following Sherlock's guideline [30], at least three elements are used through the lead thickness, with a smooth transition between the refined lead mesh and the surrounding PCB mesh. A static stress convergence study showed that refinement beyond a global PCB element edge length of 1 mm changed the extracted hotspot stress measure by less than 1%; this mesh size was therefore

Table 4.1
Layer-wise material properties for the seven layers of the M0 PCB stack-up, together with the equivalent properties used in the simplified FE models.

Entry	Material/layer	Thickness	ρ [kg/m^3]	E_{xy} [GPa]	E_z [GPa]	ν
L1	Signal copper layer (51.6% Cu)	35 µm	5463.6	60.002	60.002	–
L2	Laminate: Generic FR-4	0.36 mm	1900.0	24.804	6.634	–
L3	Signal copper layer (66.0% Cu)	35 µm	6486.0	75.770	75.770	–
L4	Laminate: Generic FR-4	0.71 mm	1900.0	24.804	6.634	–
L5	Signal copper layer (59.8% Cu)	35 µm	6045.8	68.981	68.981	–
L6	Laminate: Generic FR-4	0.36 mm	1900.0	24.804	6.634	–
L7	Signal copper layer (42.3% Cu)	35 µm	4803.3	49.818	49.818	–
Equivalent properties used in the simplified FE models						
Homogenised PCB (equiv. isotropic)		1.57 mm	2500	26.5	7.2	0.14
Copper lead (PTH)		–	8900	117.0	117.0	0.33

adopted for the PCB in all model variants. All models reproduce the experimental fixture by constraining the PCB at the four screw towers. The fixture is mounted to the shaker through the mounting holes shown in Fig. 5. The random-vibration loading is applied as base excitation using the same direction and PSD levels as in the durability tests. In linear random-vibration analysis, the response levels in the vicinity of resonances are governed by the assumed damping, because dissipation limits the resonant amplitudes. Following common practice in linear random-vibration analysis, a uniform 1% damping ratio was first adopted to define the initial baseline response. For the predictive calculations, this baseline description was replaced by mode-specific damping values, because the dominant PCB and component modes exhibited different damping levels.

4.5. Random-vibration stress response and dominant stress components

For each model variant (M1, M1_C, M3, M3_C) and each loading condition, a random-vibration analysis is performed and the frequency-domain stress response at the lead-bend hotspot is extracted in terms of the six stress components ($\sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \tau_{xy}, \tau_{yz}, \tau_{xz}$) as PSDs. Fig. 6 shows an example for case M1_C_VDR3_0.2, where the dominant responses occur in σ_{zz}, and τ_{yz} (RMS values of 134.54 MPa and 54.66 MPa, respectively), while others remain comparatively smaller. To interpret the multiaxial stress state in terms of an equivalent fatigue-driving scalar, the equivalent von Mises stress PSD $S_{eq}(f)$ is constructed from the six stress components following Preumont et al. [29] and used as input to the fatigue-damage methods (Section 3). Across models and load cases, the response is typically governed by σ_{zz} with a non-negligible contribution from τ_{yz}, while the cross-term contribution reduces the resulting equivalent stress in most cases. This decomposition is used to (i) compare the stress-driving mechanisms across model simplifications and (ii) ensure that fatigue predictions are based on the physically controlling stress components at the lead bend.

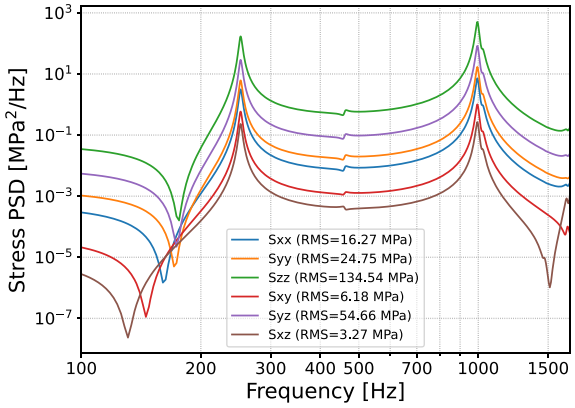


Fig. 6. Example stress PSDs at the lead-bend hotspot for case M1_C_VDR3_0.2, showing the six stress components used for subsequent equivalent-stress and fatigue calculations.

5. Results and discussion

5.1. Baseline fatigue-life predictions with 1% damping and comparison to experiments

In linear random-vibration analysis, the response levels in the vicinity of resonances are governed by the assumed damping, because dissipation limits the resonant amplitudes. A uniform 1% damping ratio is used here only to define an initial finite baseline response in the linear random-vibration workflow, before introducing the mode-specific damping description used for the predictive calculations. The stress responses obtained from the random-vibration analyses are used as input to two fatigue-life estimation approaches (Dirlik spectral method, Section 3.1, and Steinberg's three-band method, Section 3.2). For the Dirlik approach, the damage rate was evaluated using Dirlik's closed-form expression in Eq. (3), with the required spectral moments computed from the equivalent von Mises stress PSD $S_{eq}(f)$ (defined following Preumont et al. [29] and described in Section 3) and Eq. (1). For the Steinberg three-band approach, the damage rate for each configuration was computed using Eq. (6), with the zero-crossing rate N_0^+ evaluated from Eq. (5); the stress input S_{vm} was taken as the RMS equivalent von Mises stress at the hotspot (i.e., the 1σ level) derived from the same response. The fatigue parameters for the copper leads were written consistently using the positive-exponent stress–life form $N = A\,S^{-b}$, with baseline values $A = 1.7 \times 10^{17}$ and $b = 5.0$ adopted from standard high-cycle fatigue data for annealed copper. These deterministic values lie within the median statistical bounds for annealed copper wire characterised by Harlow [31], who analysed archival data to quantify the significant scatter inherent in these materials. Table 5.1 reports the resulting error factors $E = T_{\mathrm{model}}/T_{\mathrm{exp}}$ for the four PCB-level models (M1, M1_C, M3, M3_C) under the three representative loading/component combinations.

Two observations follow. First, the predicted fatigue life depends strongly on the modelling choices at both the PCB level (M1 vs. M3) and the component/lead level (baseline vs. refined "_C" variants), even under identical loading definitions. Second, and most importantly, the baseline 1% damping assumption does not yield consistent agreement with durability-test results across cases and methods: depending on the model and fatigue rule, predictions can be either overly conservative ($E \ll 1$) or non-conservative ($E \gg 1$), including extreme overprediction for some three-band cases. These discrepancies show that fatigue-life predictions derived from standard commercial workflow assumptions (1% damping, simplified leads) do not consistently agree with the durability-test results across cases and methods. The key requirement is not measurement itself, but an appropriate damping description: a finite baseline response follows from any nonzero damping assumption, whereas reliable fatigue-life prediction requires mode-specific damping values for the PCB and the critical components. Table 5.1 indicates

Table 5.1
Time-to-failure error factors for Dirlik and three-band methods, using baseline 1% damping. Error factor $E = T_{model}/T_{exp}$.

Model	Loading	Dirlik Error factor	Three band Error factor
M1_C	VDR3 : Y – 0.05	0.04	0.01
M1		**0.2**	3.3
M3_C		0.03	0.01
M3		5.4	4.5
M1_C	C14 : X – 0.2	0.04	0.02
M1		**0.6**	5.0
M3_C		0.03	0.005
M3		2.0	0.3
M1_C	VDR3 : X – 0.2	**0.3**	0.2
M1		5.5	191.4
M3_C		0.2	0.1
M3		17.3	1.2

Table 5.2
Measured natural frequencies and damping ratios for the dominant modes of VDR3, C14, and the PCB.

Mode	Measured NF (Hz)	Damping ratio (ζ)
VDR3 Mode 1	230.0	0.0389
C14 Mode 1	533.6	0.0180
PCB Mode 1	937.5	0.0101

that commercial-workflow lead idealizations (M1, M3) can produce order-of-magnitude errors under identical loading, whereas resolving the lead-bend hotspot (M1_C, M3_C) yields qualitatively different life estimates. This shows that simplified lead representations are not a reliable basis for fatigue-life assessment of heavy PTH components and that their bias cannot be removed by damping tuning alone; therefore, M1 and M3 are not considered further. The results therefore motivate revisiting the damping for the refined lead models (M1_C, M3_C) using experimentally identified, mode-dependent damping (from DIC-based displacement response and half-power bandwidth estimates) and propagating this updated damping to the stress and fatigue-life calculations.

5.2. Damping ratio identification

The large discrepancies between baseline predictions (uniform 1% damping) and the durability-test results indicate that the damping description of the PCB–component system must be calibrated. For predictive analysis, the FE model therefore requires an appropriate mode-specific damping description. Mode-specific damping ratios for the PCB, VDR3, and C14 were identified experimentally. The complete PCB–fixture assembly (with components installed) was mounted on a portable shaker through the fixture mounting holes and excited by broadband random vibration over 100–1700 Hz, matching the durability test band. Because VDR3 and C14 are small and sensitive to added mass, their motion was measured using a contactless digital image correlation (DIC) setup.

The modal damping ratios (ζ) were subsequently extracted using the half-power bandwidth method. As summarised in Table 5.2, VDR3 displays the highest damping ($\zeta \approx 3.89\%$), whereas the global PCB bending mode is lightly damped ($\zeta \approx 1.01\%$). From Table 5.2, it is evident that the dominant PCB and component modes exhibit different damping levels; accordingly, the experimentally identified damping ratios were assigned to the corresponding FE modes in the predictive analyses. A linear interpolation was applied between the distinct frequency points to define the damping across the full analysis bandwidth for all simulations (M1_C and M3_C). Fig. 7 compares the experimental FRFs (dashed lines) with the numerical predictions obtained after imposing the damping. The close agreement in peak amplitudes confirms that the experimentally calibrated damping effectively captures the energy of the physical system.

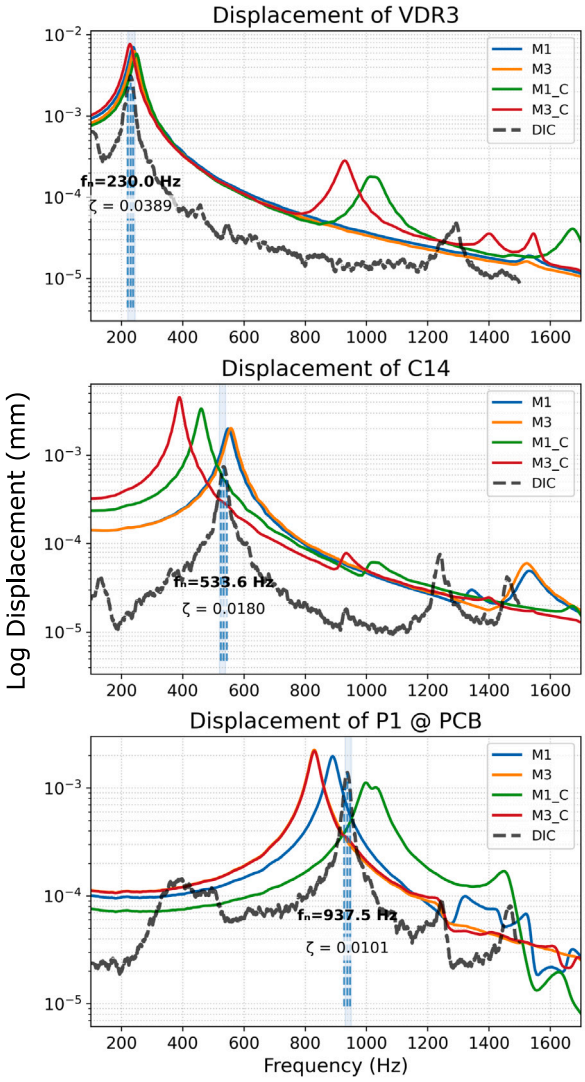


Fig. 7. Representative displacement frequency response functions (FRFs) from DIC measurements for VDR3, C14, and the PCB under random excitation, evaluated in the Y, X, and Z directions, respectively. (Note: M1 and M3 are shown for comparative purposes only).

Table 5.3
Time to failure and error factors for Dirlik and three-band methods.

Model	Loading	Dirlik time (min)	Three band Time (min)	Dirlik Error factor	Three band Error factor
M1_C	VDR3 : Y – 0.05	157.8	3283.7	**4.5**	94.9
M3_C		251.4	1553.5	7.2	44.9
M1_C	C14 : X – 0.2	11.2	132.9	**0.7**	6.8
M3_C		7.9	44.2	0.4	2.3
M1_C	VDR3 : X – 0.2	12.1	223.1	**1.3**	24.8
M3_C		13.5	107.1	1.5	11.9

5.3. Fatigue life predictions and comparison with experiments

The fatigue-life predictions for the calibrated FE models are summarised in Table 5.3. The results reveal a fundamental divergence in predictive accuracy depending on the choice of fatigue damage model and the fidelity of the lead representation.

Performance of steinberg's three-band method. The most immediate observation is that Steinberg's empirical three-band method systematically overpredicts the fatigue life of the PTH leads, regardless of the underlying FE model. Across all loading cases, error factors range from approximately 2.0 to over 30,000. Even when coupled with the most detailed geometric models (M1_C and M3_C), the method yields non-conservative predictions that deviate from experimental observations by one to two orders of magnitude in the failure-driving X–0.20 cases. This suggests that the empirical constants inherent to Steinberg's formulation—originally tuned largely for solder-joint failures in generic assemblies—are ill-suited for capturing the failure of heavy copper leads in bending.

Performance of Dirlik's spectral method. In contrast, Dirlik's spectral method demonstrates the capability to predict fatigue life with high accuracy, but *only* when the stress input is derived from a geometrically refined lead model. For the high-excitation X–0.20 cases (both VDR3 and C14), the refined models (M1_C and M3_C) combined with Dirlik's formulation result in error factors close to unity (0.4 to 1.5). This indicates that when the local stress concentration at the lead bend is explicitly resolved, the spectral method correctly captures the damage accumulation rate.

Sensitivity of the fatigue parameters (b and A) in the stress–life model. To interpret the residual mismatch observed in the long-life (low-amplitude) case, we quantify how uncertainty in the copper-lead fatigue parameters propagates into the predicted time-to-failure. The stress–life behaviour is parameterised by two Basquin constants: the exponent b, which controls the slope of the log–log S–N relation, and the coefficient A, which sets the overall life level. In practical terms, increasing A increases the predicted life nearly proportionally, while increasing b amplifies the sensitivity of life to small stress changes—a hallmark of the high-cycle regime. The explored ranges $b \in [4.0, 5.5]$ and $\log_{10} A \in [15, 18]$ were selected to (i) bracket Basquin-type fits reported for copper under fully reversed bending conditions [31] and (ii) cover the life scales typical of high-cycle copper fatigue when stresses are expressed in MPa. To quantify how uncertainty in (b, A) propagates to life predictions, an error landscape was computed by scanning $b \in [4.0, 5.5]$ and $\log_{10} A \in [15, 18]$ and evaluating the mean life ratio

$$\eta = \overline{T_{\text{model}}/T_{\text{exp}}}, \tag{9}$$

where the overbar denotes an average over the available loading/component cases using the refined lead models (M1_C and M3_C). Values $\eta > 1$ indicate non-conservative life overprediction, while $\eta < 1$ indicate conservative prediction. The resulting Dirlik and Steinberg maps are shown in Fig. 8. For example, a bending copper benchmark data set fitted with the stress–life form $N = A\,S^{-b}$ yields an exponent corresponding to $b \approx 5$ and an intercept on the order of $\log_{10} A \approx 15.8$, which lies near the centre of our sweep. A wider intercept window is retained to accommodate strong scatter due to surface condition, size effects, and local stress concentration at the lead bend. Fig. 8 summarises the sensitivity of the life error to the Basquin parameters. For Dirlik, the unity contour follows a clear diagonal trend in $(b, \log_{10} A)$ space. The baseline point lies close to this unity contour (slightly on the overprediction side), indicating that Dirlik can match the experimental life scale without large departures from the baseline parameters. This supports the conclusion from Table 5.3 that the three-band method is not robust for lead-bending dominated failures.

Global sensitivity (sobol indices). To quantify parameter influence beyond local trade-offs, we computed variance-based Sobol [34] indices using Saltelli [35] sampling. Fig. 9 shows that both parameters contribute to the output variability: the first-order indices indicate a modest direct contribution from each parameter, whereas the total-effect indices are substantially larger. The gap between S_T and S_1 for both b and $\log_{10} A$ highlights strong interaction: uncertainty in one parameter cannot be interpreted independently of the other, consistent with the diagonal compensation observed in the error landscape. This interaction-dominated behaviour is consistent with Table 5.3, where the long-life Y–0.05 case shows the largest residual mismatch even for refined lead models, reflecting the strong amplification of small parameter differences in the stress–life regime.

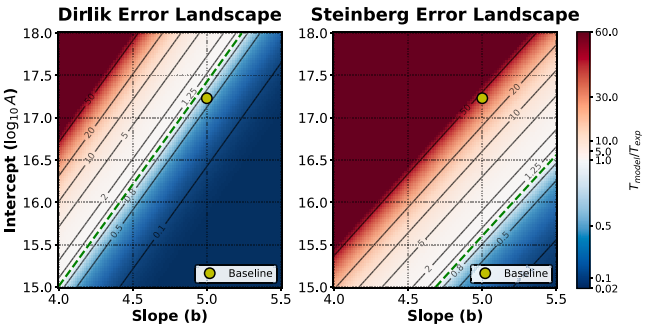


Fig. 8. Error landscapes of the mean life ratio $T_{\text{model}}/T_{\text{exp}}$ in the $(b, \log_{10} A)$ plane for the refined-lead cases (M1_C, M3_C): Dirlik (left) and Steinberg three-band (right). The dashed line marks unity and the marker denotes the baseline.

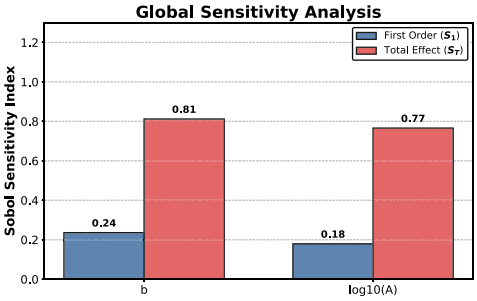


Fig. 9. Variance-based global sensitivity indices for the life-ratio metric with respect to b and $\log_{10} A$ (Dirlik-based predictions; M1_C and M3_C). First-order indices S_1 quantify the direct effect of each parameter, while total-effect indices S_T include interaction contributions.

6. Conclusions

A primary contribution of this work is a controlled broadband random-vibration time-to-failure benchmark experimental data set for lead-bending failures of two heavy plated through-hole (PTH) components (VDR3 and C14) mounted on a circulator-pump PCB. When these benchmark lives were compared with predictions from simplified PCB-level finite-element (FE) models using idealised lead representations and nominal damping, the agreement was not acceptable, demonstrating the limits of conventional modelling assumptions for this failure mode. To explain and reduce this mismatch, the study combined the experimental benchmark data with a controlled hierarchy of FE models and a mode-specific damping description. The results show that (i) locally refined representation of the lead-bend region is required to obtain credible hotspot stress responses for lead-bending dominated failures, and (ii) a parameterised mode-specific damping description is necessary to obtain credible lead-stress inputs for fatigue assessment. In contrast, simplified lead representations commonly used in fast commercial workflows can lead to overly optimistic durability predictions when lead bending governs failure. Using identical FE-derived stress inputs, the work applies both Steinberg's three-band method and Dirlik's spectral method, and observes better agreement with experiments for Dirlik in lead-bending-dominated cases. This outcome is consistent with Dirlik's PSD-based treatment of broadband stress histories, as opposed to the three-band reduction to three Gaussian stress levels.

Looking ahead, the findings reinforce the practical need for realistic representation of the lead-bend geometry together with a mode-specific damping description when assessing the durability of heavy through-hole components. In the present study, the damping values were identified from DIC measurements. The results also promote locally refined PCB-level FE models (e.g., M1_C, M3_C) as promising candidates for repeated fatigue-life evaluations in reliability studies and digital-twin applications.

CRediT authorship contribution statement

Hiren Patel: Conceptualization, Methodology, Finite element modelling, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Sergey Sorokin:** Conceptualization, Methodology, Supervision, Validation, Writing – review & editing. **Kim Martin Nielsen:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the European Commission for its support of the Marie Skłodowska Curie programme through the Horizon Europe DN APRIORI project (GA 101073551). Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union. The European Union cannot be held responsible for them.

Data availability

Data will be made available on request.

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